Unit III: Cell biology and Biomolecules

Chapter 7

Cell Cycle

Learning Objectives

The learner will be able to,

- *Outline the cell cycle and different stages in cell division.*
- *Recognise the importance of mitosis in the production of genetically identical cells.*
- *Have an insight on the significant of mitosis and meiosis.*
- *Familiarize the behaviour of chromosomes in plants and animal cells during meiosis.*



Neurons can be replaced!

Stem cells in the human brain - most neurons are in G_0 and do not divide. As neurons and neuroglia die or injured they are replaced by neural stem cells

One of the most important features of the living cells is their power to grow and divide. New cells are formed by the division of pre-existing cells. Cells increase in number by cell division. The parent cell divides and passes on genetic material to the daughter cells.

7.1 Nuclear Divisions

There are two types of nuclear division, as **mitosis** and **meiosis**. In mitosis, the daughter cells formed will have the same number of chromosomes as the parent cell, typically

Chapter Outline



7.1 Nuclear Divisions
7.2 Cell cycle
7.3 Cell Division
7.4 Difference between Mitosis and Meiosis

diploid (2n) state. Mitosis is the nuclear division that occurs when cells grow or when cells need to be replaced and when organism reproduces asexually.

In meiosis, the daughter cells contain half the number of chromosomes of the parent cell and is known as **haploid state (n)**.

Whichever division takes place, it is normally followed by division of the cytoplasm to form separate cells, called as **cytokinesis**.



Edouard Van Beneden, a Belgian cytologist, embryologist and marine biologist. He was Professor of Zoology at the University of Liège. He contributed to cytogenetics by his works on the roundworm *Ascaris*. In his work he discovered how chromosomes organized meiosis (the production of gametes).

7.2 Cell Cycle

Definition: A series of events leading to the formation of new cell is known as **cell cycle**. The series of events include several phases.

11th Botany_EM_Chapter 7.indd 166 05-01-2023 18:26:50